

DEVIXO SOLUTIONS

AI/ML INTERNSHIP PROGRAM

TASK 01

Python Programming, Data Analysis & Visualization

Dataset: Netflix Titles Dataset

Internship Domain: Artificial Intelligence & Machine Learning

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Report Sections

1. Introduction
2. Objective
3. Dataset Name & Source
4. Dataset Description
5. Dataset Size
6. Features / Columns Description
7. Methodology
8. Data Loading
9. Data Cleaning & Preprocessing
10. Exploratory Data Analysis
11. Statistical Analysis
12. Data Visualizations
13. Major Findings
14. Insights
15. Conclusion
16. GitHub Repository

1. Introduction

The Netflix Titles dataset contains information about movies and TV shows available on Netflix. It includes details such as content type, title, director, cast, country, date added, release year, rating, duration, genres, and description.

This project was completed as part of the **Devixo Solutions AI/ML Internship Program – Task 01: Python Programming, Data Analysis & Visualization**. The purpose of the project is to perform a complete data analysis workflow using Python, starting from raw dataset loading and understanding, followed by data cleaning, exploratory data analysis, statistical analysis, visualization, and insight generation.

The project provides practical experience with **Pandas, NumPy, Matplotlib, and Seaborn**. The analysis also establishes a foundation for future AI/ML tasks such as feature engineering, machine learning model development, and predictive analysis.

2. Objective

The main objective of this project is to analyze the Netflix Titles dataset and extract meaningful information from it using Python-based data analysis techniques.

The project focuses on the following objectives:

- Load and understand a real-world dataset.
 - Identify rows, columns, features, and data types.
 - Detect and handle missing values.
 - Identify and remove duplicate records.
 - Correct inappropriate data types.
 - Detect and handle invalid or inconsistent values.
 - Perform exploratory data analysis (EDA).
 - Calculate statistical measures such as mean, median, mode, minimum, maximum, standard deviation, and variance.
 - Analyze relationships between numerical variables.
 - Create meaningful data visualizations.
 - Identify trends, patterns, distributions, and outliers.
 - Generate at least 10 meaningful insights from the dataset.
 - Present the complete analysis professionally through documentation, a Streamlit dashboard, and a GitHub repository.
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3. Dataset Name & Source

Dataset Name: Netflix Titles Dataset

Dataset Source: Kaggle

Dataset Size: 8,807 records and 12 columns.

Dataset Type: Real-world entertainment/content dataset containing both categorical and numerical information.

4. Dataset Description

The **Netflix Titles Dataset** contains information about movies and TV shows available on Netflix. The dataset provides details about the content itself, including its title, type, director, cast, country of production, date added to Netflix, release year, content rating, duration, genres, and description.

The dataset contains both **categorical and numerical variables**, making it suitable for exploratory data analysis, statistical analysis, and data visualization.

The dataset originally contained missing values in several columns, particularly **director**, **cast**, and **country**. These issues were identified and handled during the data-cleaning phase.

5. Dataset Size

The selected Netflix dataset contains:

- **Total Rows:** 8,807
- **Total Columns:** 12
- **Total Records:** 8,807 Netflix titles
- **Content Types:** Movies and TV Shows

The dataset contains **6,131 Movies** and **2,676 TV Shows**.

Netflix Content Type

Number of movies and TV shows in the selected dataset.

02,0004,0006,0008,000MovieTV Show

6. Features / Columns Description

Column	Data Type	Description
show_id	String	Unique identifier assigned to each Netflix title.
type	Categorical	Indicates whether the content is a Movie or TV Show.
title	String	Name/title of the movie or TV show.
director	String	Name of the director associated with the title.
cast	String	Actors and performers appearing in the title.
country	String	Country or countries associated with production.
date_added	Date/Time	Date when the title was added to Netflix.
release_year	Integer	Original release year of the movie or TV show.
rating	Categorical	Content rating such as TV-MA, TV-14, PG, R, etc.
duration	String	Duration of a movie in minutes or number of seasons for a TV Show.

`listed_in` Categorical Genres/categories associated with the title.

`description` String Short description or summary of the title.

Column Classification

Numerical:

- `release_year`

Categorical:

- `type`
- `rating`
- `country`
- `listed_in`

Date/Time:

- `date_added`

Text/String:

- `show_id`
- `title`
- `director`
- `cast`
- `duration`
- `description`

Note: `duration` is stored as text because it contains both values such as "**90 min**" for movies and "**2 Seasons**" for TV shows.

7. Methodology

The project followed a complete data analysis workflow, beginning with dataset selection and ending with insights and conclusions. The major stages of the methodology were:

Dataset Selection → Data Loading → Data Understanding → Data Cleaning → EDA → Statistical Analysis → Visualization → Insights → Conclusion → Dashboard → Documentation

7.1 Dataset Loading

The Netflix dataset was loaded into a Pandas DataFrame using Python. The initial dataset was inspected to verify that the data had been loaded correctly.

7.2 Data Understanding

The structure of the dataset was examined using:

- First and last records
- Random samples
- Dataset shape
- Column names
- Data types
- Non-null values
- Missing values
- Basic statistical information

The dataset contained **8,807 rows and 12 columns**.

7.3 Data Cleaning & Preprocessing

The dataset was checked for common data-quality problems, including:

- Missing values
- Duplicate records
- Incorrect data types
- Inconsistent values
- Invalid values

Missing values were handled according to the nature of each column. Duplicate records were checked and no duplicate records were found.

The `date_added` column was converted from string format to a proper date/time data type.

Three records were also identified where movie duration values (`74 min`, `84 min`, and `66 min`) had incorrectly appeared in the `rating` column. These inconsistent values were corrected

by moving the duration information to the appropriate **duration** field and handling the corresponding rating values.

7.4 Exploratory Data Analysis

EDA was performed to understand the distribution and characteristics of Netflix content. The analysis examined:

- Movies vs TV Shows
- Content ratings
- Countries
- Release years
- Content distribution
- Common categories
- Numerical statistics
- Relationships between variables

7.5 Statistical Analysis

Statistical measures were calculated to understand the numerical characteristics of the dataset, including:

- Mean
- Median
- Mode
- Minimum
- Maximum
- Standard deviation
- Variance
- Correlation analysis

7.6 Data Visualization

Multiple visualization techniques were used to communicate patterns and relationships in the dataset, including:

- Histogram
- Bar chart
- Pie chart
- Scatter plot
- Line chart
- Box plot
- Correlation heatmap
- Count plot
- Pair plot

- Violin plot

7.7 Insight Generation

The results of the analysis were interpreted to identify meaningful patterns, trends, distributions, relationships, and unusual observations.

7.8 Interactive Dashboard

A Streamlit dashboard was also developed as an optional extension. The dashboard provides KPI summaries, content-type and rating filters, charts, country analysis, and a preview of the filtered dataset.

8. Data Loading

The dataset was loaded using the **Pandas** library. The CSV file was stored inside the project's `dataset` directory and imported into a Pandas DataFrame.

The initial inspection confirmed that the dataset contained:

8,807 rows × 12 columns

The first five records, last five records, and sample records were examined to understand the structure and contents of the dataset.

Initial Dataset Structure

Property	Result
Rows	8,807
Columns	12
Numerical columns	1

Date/Time columns 1

Text/Categorical columns 10

Duplicate records 0

Data Loading Tools

The following Python libraries were used throughout the project:

- **Pandas** — data loading, cleaning, manipulation and analysis
 - **NumPy** — numerical operations
 - **Matplotlib** — data visualization
 - **Seaborn** — statistical visualization
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9. Data Cleaning & Preprocessing

Data cleaning was performed to improve the quality, consistency, and reliability of the Netflix dataset before conducting detailed analysis. The dataset was examined for missing values, duplicate records, incorrect data types, and inconsistent values.

9.1 Missing Values

The initial missing-value analysis identified missing data in four columns:

Column	Missing Values	Percentage
director	2,634	29.91%
cast	825	9.37%

country	831	9.44%
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date_added	10	0.11%
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rating	4	0.05%
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duration	3	0.03%
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The remaining columns did not contain missing values.

The missing values were handled according to the characteristics of each column. For text-based fields such as `director`, `cast`, and `country`, missing information was not replaced with arbitrary values because doing so could introduce misleading information. Where appropriate for analysis, missing categorical information was represented as **Unknown**.

For `date_added`, the missing values were retained as missing because there was no reliable information available to determine the actual Netflix addition date.

9.2 Duplicate Records

Duplicate records were checked using Pandas.

The analysis produced:

Duplicate records: 0

Therefore, no duplicate rows needed to be removed from the dataset.

This confirmed that every row represented a unique record in the dataset.

9.3 Data Type Correction

The original dataset contained several columns stored as strings. The `date_added` column was originally stored as a string and was converted into a proper date/time format.

Before:

`date_added → string`

After:

`date_added → datetime`

The `release_year` column was already correctly stored as an integer.

9.4 Inconsistent Rating Values

During the analysis of the `rating` column, three records were identified with incorrect values:

Title	Incorrect Rating	Actual Duration
Louis C.K. 2017	74 min	74 min
Louis C.K.: Hilarious	84 min	84 min
Louis C.K.: Live at the Comedy Store	66 min	66 min

Values such as `74 min`, `84 min`, and `66 min` are durations rather than content ratings. These records had their duration information corrected and their rating values handled appropriately.

This cleaning step was important because leaving these values in the `rating` column would have produced inaccurate rating-frequency analysis and visualizations.

9.5 Duration Column

The `duration` column contains two different types of information:

- Movies → duration in minutes, e.g. 90 min
- TV Shows → number of seasons, e.g. 2 Seasons

Therefore, the column was retained as a categorical/text field rather than treating all values as a single numerical measurement.

For example:

90 min → Movie duration

2 Seasons → TV Show duration

This distinction was considered during the EDA and visualization stages.

9.6 Cleaning Verification

After preprocessing, the dataset was verified again.

Verification	Result
Rows	8,807
Columns	12
Duplicate records	0
release_year type	Integer
date_added type	Date/Time
Major invalid rating values	Corrected

The cleaned dataset was then used for exploratory data analysis and statistical analysis.

10. Exploratory Data Analysis

After cleaning the dataset, exploratory data analysis was performed to understand the characteristics, distributions, and patterns within the Netflix content.

The EDA focused on content type, ratings, countries, release years, and statistical properties of the numerical data.

11. Statistical Analysis

Statistical analysis was performed to quantitatively understand the numerical characteristics and distributions present in the Netflix dataset. The `release_year` column was selected as the primary numerical variable because it contains numerical values representing the original release year of each title.

11.1 Mean

The mean release year was calculated as:

Mean = 2014.18

This indicates that the average release year of the titles in the dataset is approximately **2014**.

11.2 Median

The median release year was:

Median = 2017

This means that half of the titles have a release year below or equal to 2017, while the other half have a release year above or equal to 2017.

The median is particularly useful because extreme older release years can affect the mean.

11.3 Mode

The most frequently occurring release year was:

Mode = 2018

This indicates that **2018** was the most common release year among the titles in the dataset.

11.4 Minimum and Maximum

The minimum release year was:

1925

The maximum release year was:

2021

Therefore, the dataset contains titles released across a very wide historical period.

11.5 Standard Deviation

The standard deviation of the release year was:

8.82

This indicates the amount of variation in release years around the mean. A higher standard deviation would indicate greater variation, while a lower value would indicate that observations are more concentrated around the average.

11.6 Variance

The variance was calculated as:

77.78

Variance represents the squared measure of dispersion and supports the interpretation provided by the standard deviation.

11.7 Correlation Analysis

Correlation analysis was performed on suitable numerical variables to identify linear relationships between numerical features.

Since the cleaned Netflix dataset contains only one primary numerical feature, **release_year**, there are limited opportunities for meaningful numerical-to-numerical correlation analysis.

Therefore, categorical variables such as **type**, **rating**, and **country** were analyzed separately through frequency-based analysis and visualizations rather than applying numerical correlation to them.

11.8 Statistical Summary

Statistical Measure	Release Year
Mean	2014.18
Median	2017
Mode	2018
Minimum	1925
Maximum	2021
Standard Deviation	8.82
Variance	77.78

Statistical Interpretation

The statistical analysis indicates that Netflix titles in the dataset are concentrated around relatively recent release years, with a median of 2017 and mode of 2018. However, the presence of much older titles, including content from 1925, lowers the mean to 2014.18.

This difference between the mean and median suggests that older titles have an influence on the overall distribution.

12. Data Visualizations

A total of **10 different visualizations** were created to analyze the Netflix dataset from different perspectives. Each visualization was selected according to the type of information it represents and was used to identify meaningful patterns and trends.

12.1 Histogram — Release Year Distribution

The histogram was used to examine the distribution of Netflix titles across different release years.

Interpretation:

The visualization shows how Netflix titles are distributed over time. A larger concentration of titles can be observed in recent release years, indicating that the dataset contains considerably more modern content than older content. The presence of older titles creates a longer distribution toward earlier years.

12.2 Bar Chart — Movies vs TV Shows

The bar chart compares the number of Movies and TV Shows in the dataset.

Interpretation:

Movies are significantly more numerous than TV Shows. The dataset contains **6,131 Movies** compared with **2,676 TV Shows**, demonstrating that Movies represent the majority of Netflix titles in the selected dataset.

12.3 Pie Chart — Content Type Distribution

The pie chart represents the proportion of Movies and TV Shows within the complete dataset.

Interpretation:

Movies account for approximately **69.62%** of the dataset, while TV Shows account for approximately **30.38%**. Therefore, roughly two-thirds of the available records are Movies.

12.4 Scatter Plot — Release Year Analysis

A scatter plot was used to visualize the distribution of titles across release years.

Interpretation:

The scatter visualization provides a view of how titles are distributed across the release-year range. It also helps identify the concentration of titles in more recent years and the presence of older content.

12.5 Line Chart — Titles by Release Year

The line chart was used to examine changes in the number of Netflix titles across release years.

Interpretation:

The visualization highlights how the number of titles changes across different release years. The pattern demonstrates that Netflix's catalog contains a stronger concentration of titles from recent years compared with earlier periods.

12.6 Box Plot — Release Year

The box plot was used to examine the distribution and spread of release years and identify potential outliers.

Interpretation:

The box plot demonstrates the central distribution of release years and highlights unusually old release years as potential outliers. The presence of older titles, particularly titles from the early part of the dataset's range, contributes to the extended distribution.

12.7 Correlation Heatmap

A correlation heatmap was created for suitable numerical variables.

Interpretation:

The correlation analysis was limited because the dataset contains only one primary numerical variable, `release_year`. Therefore, the heatmap does not provide extensive numerical relationships between independent variables. Categorical variables were instead analyzed using frequency-based visualizations.

12.8 Count Plot — Content Ratings

The count plot was used to display the frequency of different content ratings.

Interpretation:

The visualization shows that **TV-MA** is the most frequently occurring rating, followed by **TV-14** and **TV-PG**. This indicates a strong representation of mature and teenage-oriented content within the dataset.

12.9 Pair Plot

A pair plot was used to examine relationships among available numerical variables.

Interpretation:

Because the cleaned dataset contains limited numerical features, the pair plot provides limited multi-variable relationship analysis. Nevertheless, it provides a visual overview of the available numerical distribution and confirms the limited numerical dimensionality of this dataset.

12.10 Violin Plot — Release Year Distribution

The violin plot was used to examine the distribution and density of release years.

Interpretation:

The violin plot provides a visual representation of where release years are concentrated. The distribution shows a stronger concentration around recent years, while older titles appear less frequently.

Visualization Summary

The 10 visualizations provided different perspectives on the Netflix dataset:

1. **Histogram** → release-year distribution
2. **Bar Chart** → Movies vs TV Shows
3. **Pie Chart** → content-type proportions
4. **Scatter Plot** → release-year distribution
5. **Line Chart** → release-year trend
6. **Box Plot** → spread and potential outliers
7. **Heatmap** → numerical correlation

8. **Count Plot** → rating frequencies
9. **Pair Plot** → numerical-variable relationships
10. **Violin Plot** → release-year density

13. Major Findings

The analysis of the Netflix Titles dataset produced several important findings about the distribution and characteristics of Netflix content.

1. **Movies dominate the dataset.**
Movies account for 6,131 titles, while TV Shows account for 2,676 titles.
2. **Movies represent approximately 69.62% of the dataset.**
TV Shows represent approximately 30.38%, showing a clear difference between the two content types.
3. **TV-MA is the most common content rating.**
There are 3,214 titles with a TV-MA rating, making it the most frequently occurring rating.
4. **TV-14 is the second most common rating.**
A total of 2,160 titles have a TV-14 rating.
5. **The United States has the highest number of titles.**
The United States is associated with 2,818 titles, followed by India with 972 titles.
6. **The dataset covers a very wide release period.**
Titles range from 1925 to 2021, demonstrating that Netflix includes both classic and modern content.
7. **Recent content is strongly represented.**
The median release year is 2017, indicating that a large portion of the dataset consists of relatively recent titles.
8. **2018 is the most frequent release year.**
The mode of the `release_year` variable is 2018.
9. **Older titles influence the average release year.**
The mean release year is 2014.18, which is lower than the median of 2017 because the dataset contains considerably older titles.
10. **The dataset contains significant variation in release years.**
The standard deviation is approximately 8.82 years.
11. **Missing information is concentrated in specific columns.**
The `director`, `cast`, and `country` columns contain considerably more missing values than the other columns.
12. **The dataset contains both Movies and TV Shows with different duration formats.**
Movie durations are generally represented in minutes, while TV Show durations are represented in seasons.

13. The dataset contains inconsistent records that required cleaning.

Three movie records contained duration values such as 74 min, 84 min, and 66 min in the rating column.

14. Insights & Findings

Based on the exploratory and statistical analysis, the following insights were identified:

Insight 1 — Movies are the dominant content type

Movies make up approximately **69.62%** of all records, showing that Movies form the majority of the selected Netflix catalog.

Insight 2 — TV Shows have a smaller representation

TV Shows account for approximately **30.38%** of the dataset, which is less than half the number of Movie records.

Insight 3 — Mature content is highly represented

TV-MA is the most common rating with **3,214 titles**, indicating a substantial amount of mature-oriented content.

Insight 4 — TV-14 is also highly frequent

TV-14 appears on **2,160 titles**, making it the second most common rating.

Insight 5 — United States dominates country representation

The United States appears with **2,818 titles**, considerably more than any other individual country.

Insight 6 — India is the second major contributor

India appears with **972 titles**, making it the second most represented country in the dataset.

Insight 7 — Netflix contains content from different eras

The release years range from **1925 to 2021**, demonstrating a broad historical range of content.

Insight 8 — Recent titles dominate the central distribution

The median release year is **2017**, suggesting that the typical title in the dataset is relatively recent.

Insight 9 — 2018 is the most common release year

The mode is **2018**, meaning this year occurs more frequently than any other release year in the dataset.

Insight 10 — Older content affects the average

The mean release year (**2014.18**) is lower than the median (**2017**), indicating that older titles have an influence on the overall average.

Insight 11 — Release years show variation

The standard deviation of **8.82** indicates noticeable variation in release years around the mean.

Insight 12 — Missing data is not evenly distributed

The **director** column has the largest amount of missing information, with **2,634 missing values**, while columns such as **show_id**, **title**, **type**, and **description** contain no missing values.

Insight 13 — Data quality problems can affect analysis

The three incorrect rating records demonstrate why data validation and cleaning are important before statistical analysis and visualization.

15. Final Conclusion

The Netflix Titles dataset was successfully analyzed as part of the **Devixo Solutions AI/ML Internship Program Task 01**. The project involved loading and understanding a real-world dataset containing **8,807 records and 12 columns**, followed by data cleaning, exploratory data analysis, statistical analysis, visualization, and insight generation.

During data preprocessing, missing values, duplicate records, incorrect data types, and inconsistent values were examined. The **date_added** column was converted into a date/time format, duplicate records were verified, and incorrect duration values found in the rating column were corrected. Exploratory and statistical analysis revealed that Movies represent the majority of the dataset, TV-MA is the most common rating, and the United States is the most represented country.

The project provided practical experience with **Python, Pandas, NumPy, Matplotlib, Seaborn, and Streamlit**. The analysis demonstrates a complete data-analysis workflow and provides a strong foundation for future AI/ML tasks involving feature engineering, machine learning models, predictive analysis, and model evaluation.

16. GitHub Repository

The complete project has been organized into a structured GitHub repository containing the analysis notebook, dataset, Streamlit dashboard, documentation, and supporting project files.

GitHub Repository:

[\[https://github.com/shamii-101/DEVIOX_SOLUTION.git\]](https://github.com/shamii-101/DEVIOX_SOLUTION.git)

Repository Name: [Netflix-Data-Analysis](#)
